

How to prepare a scientific poster?

Module: Earth System Modelling

Josephin Kroll, 10.01.24

Purpose of a poster

- General academic setting:
 - Present preliminary results to spark discussions
 - Present current publications to create interest
 - Connect with researchers in your field
- In our seminar:
 - Present results of your analysis with the water balance model
 - Share knowledge with your fellow students
 - Induce curiosity and interest to start discussions

Purpose of a poster

- General academic setting:
 - Present preliminary results to spark discussions

It's all about getting attention and your topic being remembered!

- Present results of your analysis with the water balance model
- Share knowledge with your fellow students
- Induce curiosity and interest to start discussions

**How to make your topic being
remembered?**

How to make your topic being remembered?

1. Title

- **Formulate an intriguing question regarding your topic.**
 - What will happen to heatwave intensity if we overstep the 1.5°C of global warming?
- **State the main finding your poster illustrates.**
 - Heatwave intensity increases are halved when limiting global warming to 1.5°C.
- **Use a famous quote applied to your topic.**
 - You shall not pass: Reducing heatwave intensity increases by limiting global warming to 1.5°C.

How to make your topic being remembered?

2. Introduction

1. Give a short wrap-up of the state of the art research.
 - What is the current knowledge base for the components of your topic.
 - Which views exist regarding the topic, are they opposing?
 - Which questions do result from this?
 - Tailor this part towards your topic as explanation for the choice of it.
2. State and shortly explain your topic as research questions and/or hypotheses.

Don't introduce a topic your not working on.



How to make your topic being remembered?

3. Data & Methods

- In Research:
 - Used data sets
 - Information on calculations, statistics, wording, etc.
 - Explanation of used models
 - Important terminology

How to make your topic being remembered?

4. Results

- Formulate the main results yielded from your literature/data analysis.
- Take care that your main findings match your research question, hypothesis and title.
- Use illustrations.
 - Diagrams, pictograms, pictures, schemes, maps, ...
 - Put meaningful figure captions: Short description and the finding this figure illustrates.

What do you see?

4. Results



- Larger points accumulate at high x values.
- Four groups (=colors) can be detected.

With increasing X values, Y values increase.

- Most data points are in the center of the distribution.
- The largest data point seems to be an outlier.

How to make your topic being remembered?

5. Conclusion and discussion (separate or combined)

- Could you answer your hypothesis/research questions clearly? What is the answer?
- Do your results agree or disagree with previous related studies?
 - Suggest reasons for disagreement.
- Did new questions come out of your analysis?
- Do your results highlight the need/urgency for something?
- Which limitations exist for your analysis? Consider your methods, the data used in terms of length, uncertainty, resolution, ...
- Information on what is planned next, how could the study be continued/completed

How to make your topic being remembered?

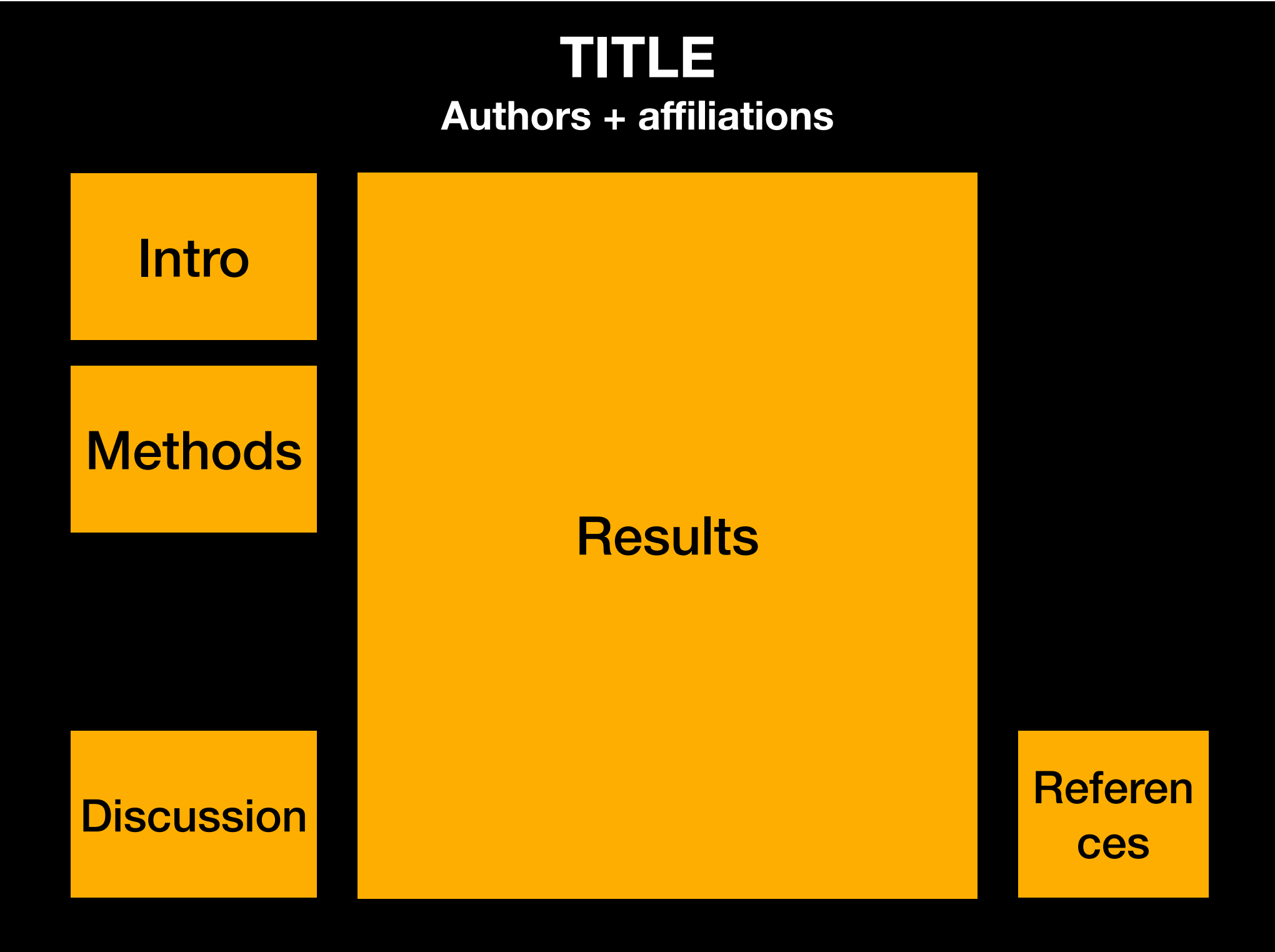
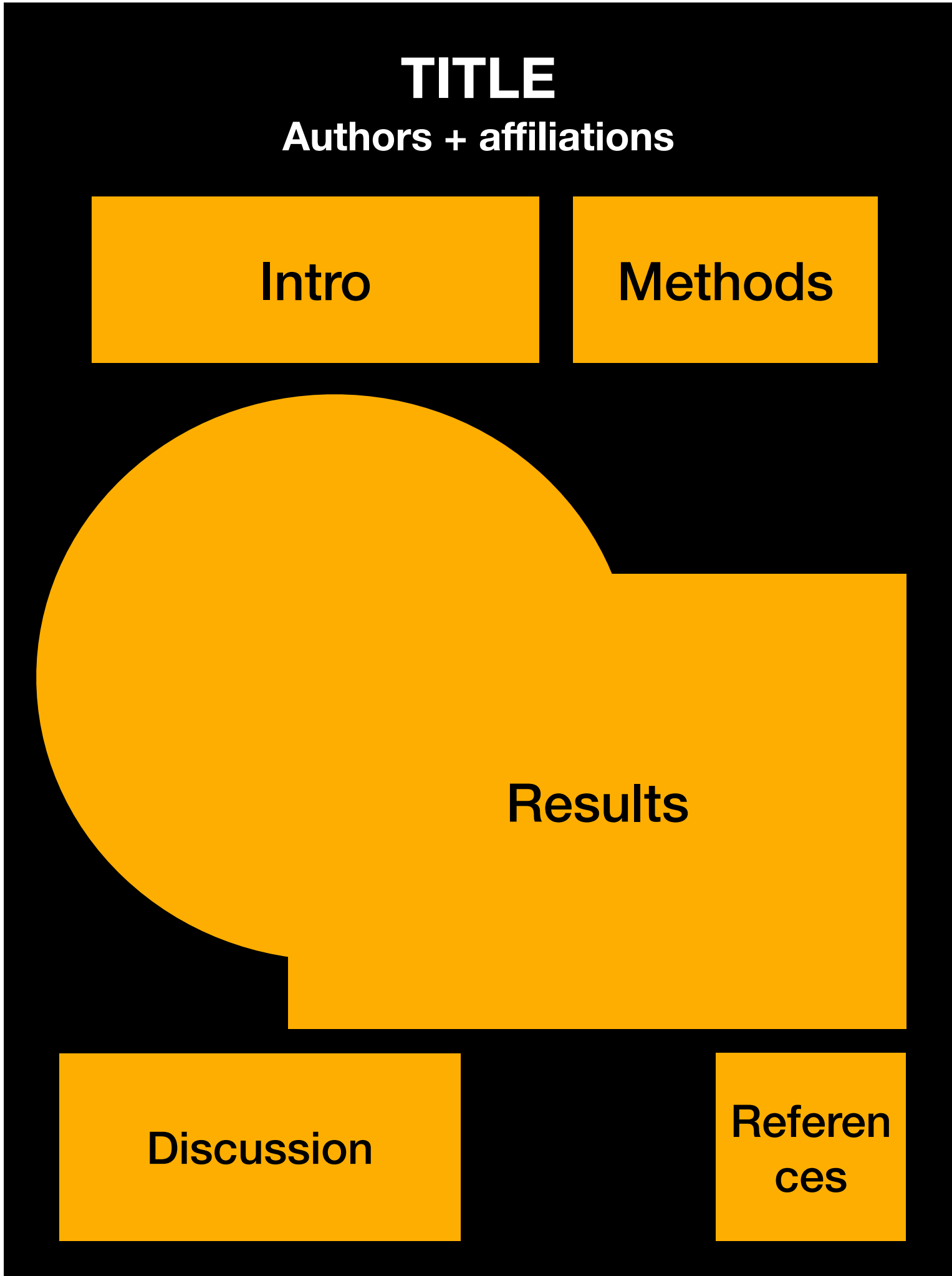
6. References

- You include and use findings from other people. Acknowledge this by putting references to their work.
- Save space by linking those with a QR code.

How to get attention?

How to get attention?

General structure of a poster



How to get attention?

General structure of a poster

- Use structuring elements: boxes, headings, highlighting.
- Avoid block text to create a lighter impression —> go for bullet points.
- Consider numbering for guidance of the reader.
- Be creative.
 - Leave a blank space to collect post-its from viewers with suggestions/ thoughts
 - Adding elements where you can pull/unfold something to gain more information.

How to get attention?

Make it look nice.

Design your poster

- Use eye-catchers: Play with the size of your figures.
- Divide the space appealingly: Try how it looks if you center important objects.

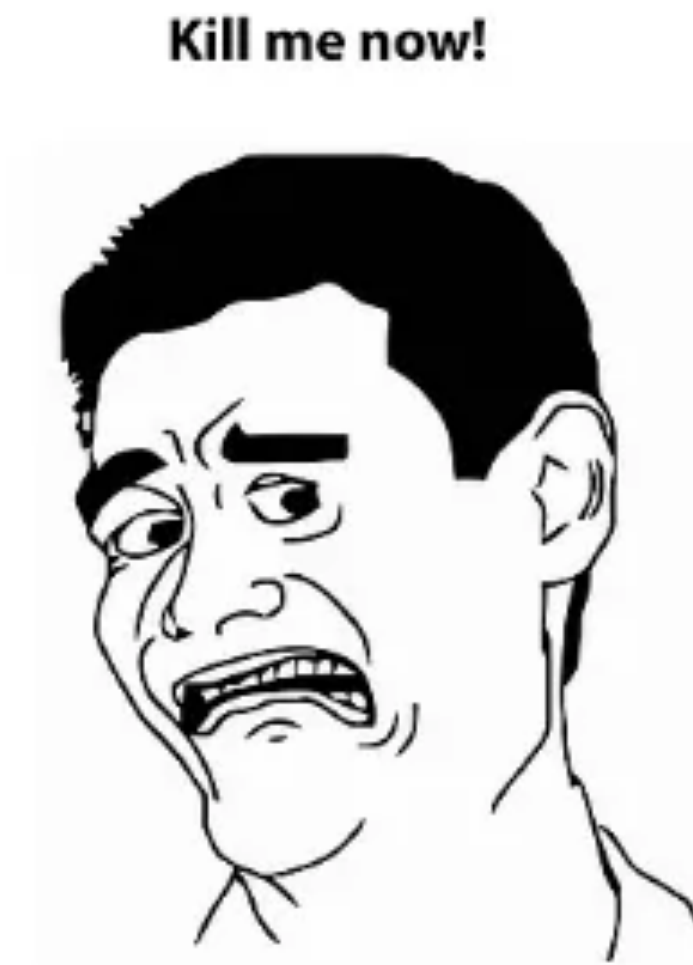


How to get attention?

Make it look nice.

Design your poster

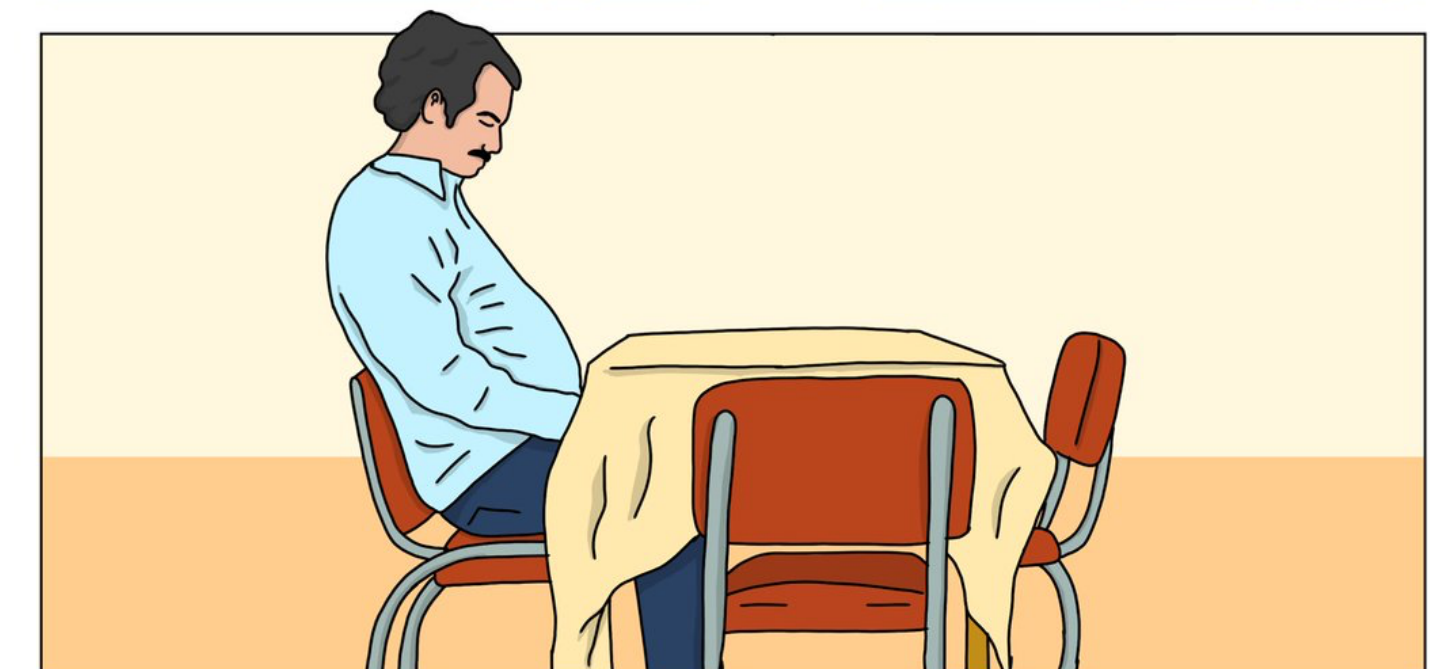
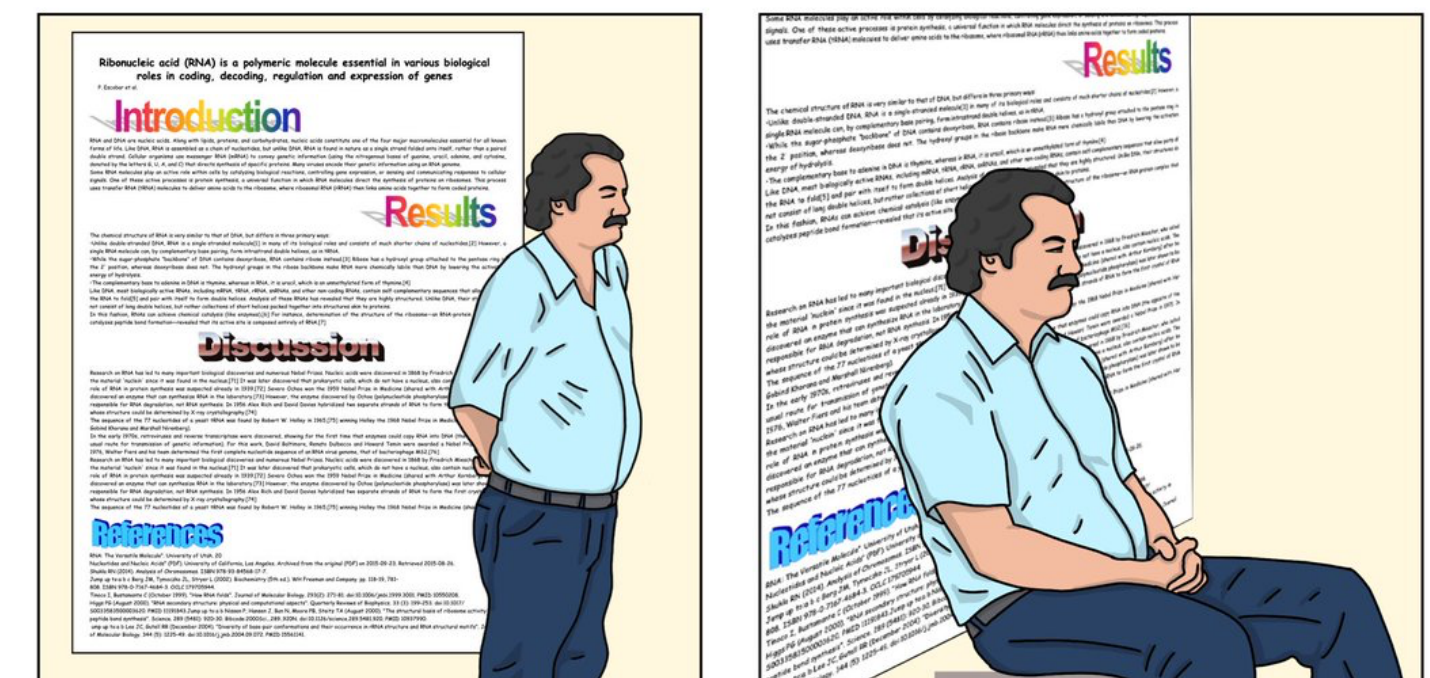
- Use eye-catchers
- Divide the space appealingly
- Make it readable
 - Textsize-wise
 - Font-wise
 - Color-wise
 - Amount-wise



Kill me now!



When you used **Comic Sans** and **WordArt**, and no one visited your poster



How to get attention?

- Imagine yourself in a hall with 500 posters at 3pm, after hearing three complex 1h-talks that you tried to follow. You missed the coffee break.
- What would make you approach a poster (design-wise)?
 - Simplicity
 - Structure
 - Colours: appealing/balanced vs. loud/dazzling/glaring
 - Space

Help

- Content:
 - Google scholar, Web of Science, Carbon Brief, IPCC + Interactive Atlas, google time lapse
 - Find related literature: connected papers
- Design:
 - Explore designs on Etsy, Pinterest, Google, etc. “scientific poster design”
 - Search for general design ideas, color combinations, etc.
 - Create illustrations/diagrams using PP, Excel, R, chatGPT, Canva, etc.

Example 1



Towards a more observation-based representation of vegetation in weather forecasting

Melissa Ruiz-Vásquez^{1,2,3,*}, Sungmin O⁴, Alexander Brenning², Gianpaolo Balsamo³, Souhail Boussetta³, Gabriele Arduini³ and René Orth¹

1. Introduction

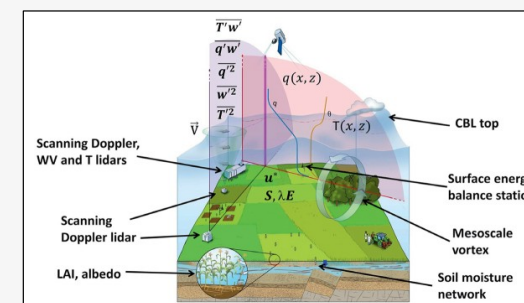


Fig. 1 Land-atmosphere interactions and their monitoring systems
(From Wulfmeyer et al., 2018)

•Vegetation anomalies impact latent heat flux through evapotranspiration and hence potentially temperature and precipitation.

•Despite the developments in Earth observational systems and the availability of land surface information, most weather forecast models do not explicitly consider the state of the vegetation in their formulation.

2. Research question

What is the impact of (i) land cover type, (ii) Leaf Area Index variability and (iii) soil and vegetation related parameters on soil moisture and latent heat flux within a land surface model?

3. Methodology

First experiment: uncoupled simulations

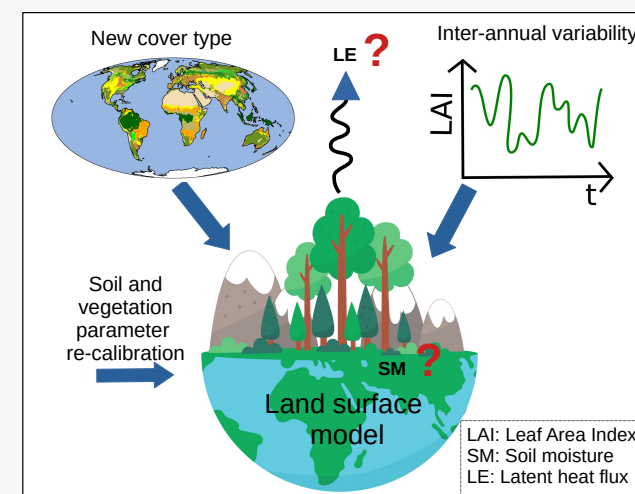


Fig. 2 Schematics of the uncoupled modelling experiments

•We investigate the role of inter-annual variability of the vegetation and updated cover type.

•We re-calibrate the following land surface model parameters based on soil moisture and latent heat flux reference values:

- Hydraulic conductivity
- Skin conductivity
- Total soil depth
- Minimal stomatal resistance
- Soil moisture stress function

4. Preliminary results

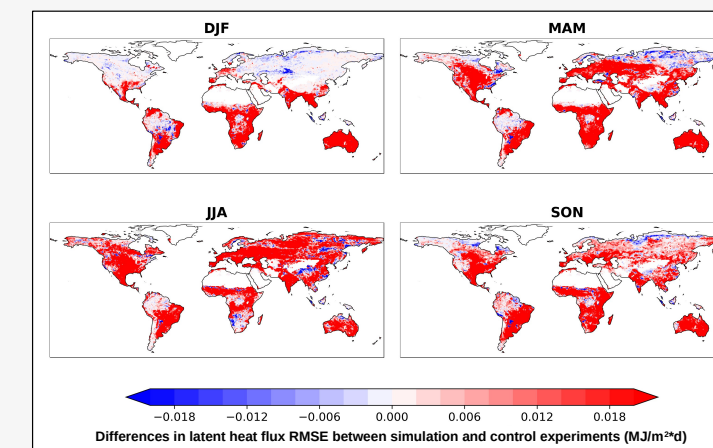


Fig. 3 Model performance in terms of latent heat flux

The model performance in terms of latent heat flux is degraded (red) with the new land surface scheme configuration.

Reference data: FLUXCOM

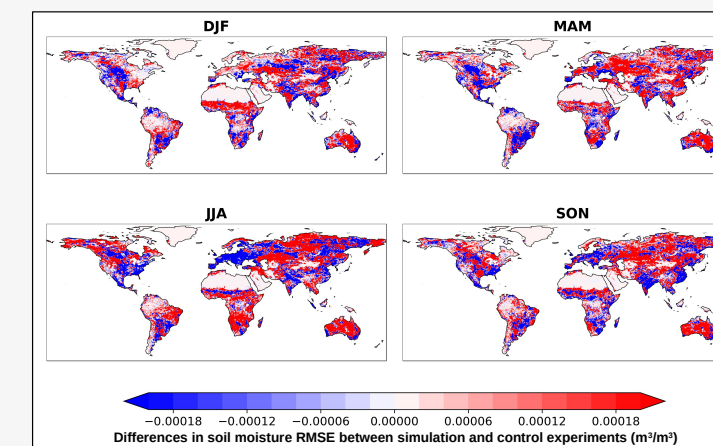


Fig. 4 Model performance in terms of soil moisture

There are some areas with improved model performance (blue) in terms of soil moisture with the new land surface configuration. Still, there are other areas with worse model performance (red).

Reference data: SoMo.ml

5. Take home message

- Latent heat flux and soil moisture respond differently to the land surface changes.
- Updating the input data used in the land surface model does not necessarily improve the accuracy of the output.

6. Outlook

Second experiment: coupled simulations

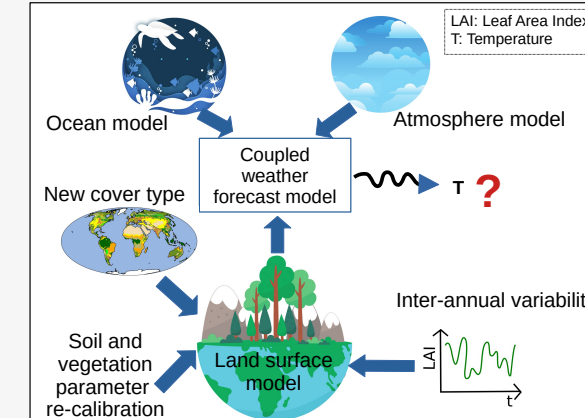


Fig. 5 Schematics of the coupled modelling experiments

•We will run global coupled weather forecasts using the best land surface configuration of the uncoupled simulations.

•We will quantify the temperature forecast errors with the new land surface scheme.

Further analysis: under extreme events

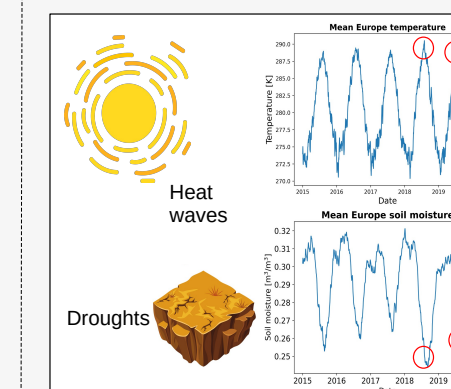


Fig. 6 Schematics of the model performance with the new land surface scheme configuration under extreme events

What is the impact of inter-annual variability of vegetation on soil moisture, latent heat flux and temperature forecasts during extreme events?

Reference

Wulfmeyer, V., Turner, D. D., Baker, B., Banta, R., Behrendt, et al. (2018). A New Research Approach for Observing and Characterizing Land–Atmosphere Feedback. Bulletin of the American Meteorological Society, 99(8), 1639–1667. <https://doi.org/10.1175/BAMS-D-17-0009.1>

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Example 2

Atmospheric Circulations and Drought Conditions in British Catchments: Highlighting the Role of the East Atlantic Pattern

Harry West, Neill Quinn & Michael Horswell

School of Architecture and Environment, University of the West of England, Bristol, UK | harry.west@uwe.ac.uk

UWE Bristol University of the West of England



1. INTRODUCTION

The spatiality of drought events is complex in Britain, as is the propagation of rainfall to flows within catchments (Barker et al. 2016; Tanguy et al. 2021). Distinctive differences in both rainfall and flow regimes have been found between the north-western and south-eastern regions which are related to a range of climatological and hydrological variables. Spatial patterns in the propagation of drought is further complicated as it is influenced by both climate and catchment characteristics (Van Loon & Laaha 2015). Droughts are inherently driven by climatic processes, and for Britain and most of NW Europe, the North Atlantic Oscillation (NAO) atmospheric-oceanic circulation has long been identified as the leading mode of climate variability (Hurrell et al. 2003) affecting rainfall across Britain (West et al. 2019). The NAO is defined by two meridional dipoles—the Icelandic Low and Azores High/Anticyclone. When the sea level pressure difference between these two locations is greater than average the NAO is said to be in a positive phase (NAO+), whilst a weaker than average SLP difference represents a negative phase (NAO-).

The East Atlantic Pattern (EA) has also been acknowledged to have an impact on climate (West et al. 2021; 2022b) and has been referred to as a southward shifted NAO-like pattern. Studies have noted that the phase and magnitude of the EA also influences rainfall spatial distribution and volume across Europe, and consequently by using a combination of the NAO and EA we may be able to describe climate variability more accurately (Moore et al. 2013; Comas-Bru & McDermott, 2014).

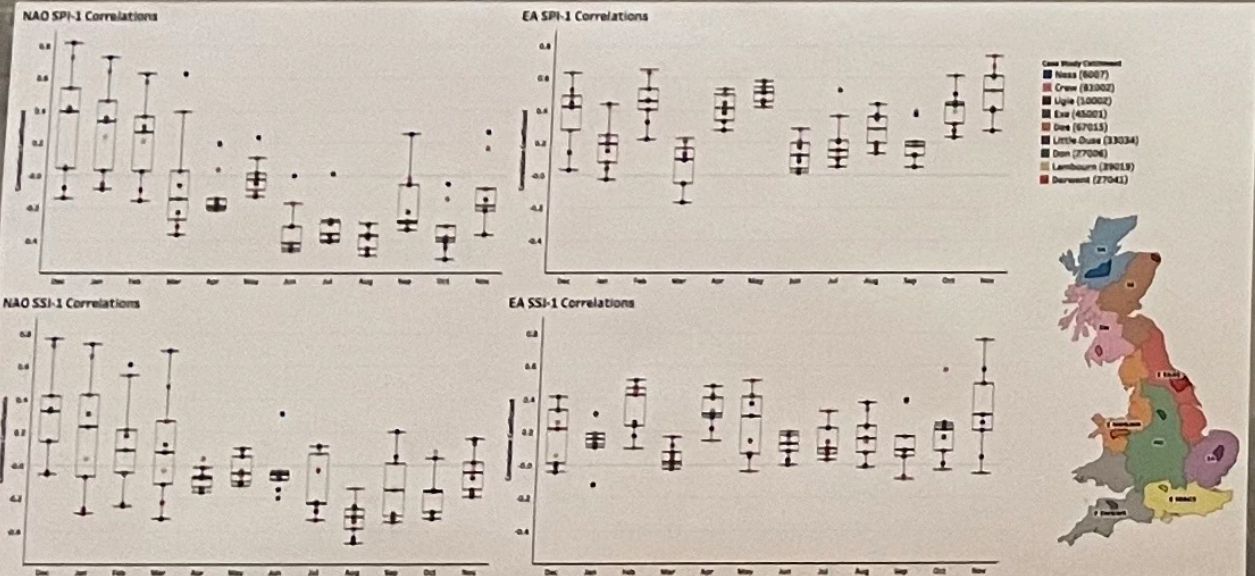
The above highlights the complex interplay between climate and hydrological systems, which manifests in spatiotemporal variability in rainfall, flow responses, and subsequent drought characteristics and propagation, generally along a N/S or NW/SE gradient in Britain. This study aims to bring together understandings of the influence of the NAO and EA on rainfall distribution and magnitude, and the variable nature of meteorological to hydrological drought (rainfall-stream/flow) propagation.

To assess the general relationship between the NAO/EA and rainfall and streamflow, correlation analysis was undertaken for 291 catchments across GB. Rainfall is represented by the Standardised Precipitation Index (SPI) and flow by the Standardised Streamflow Index (SSI) – both with a one month accumulation period (Tanguy et al. 2017; Barker et al. 2018).

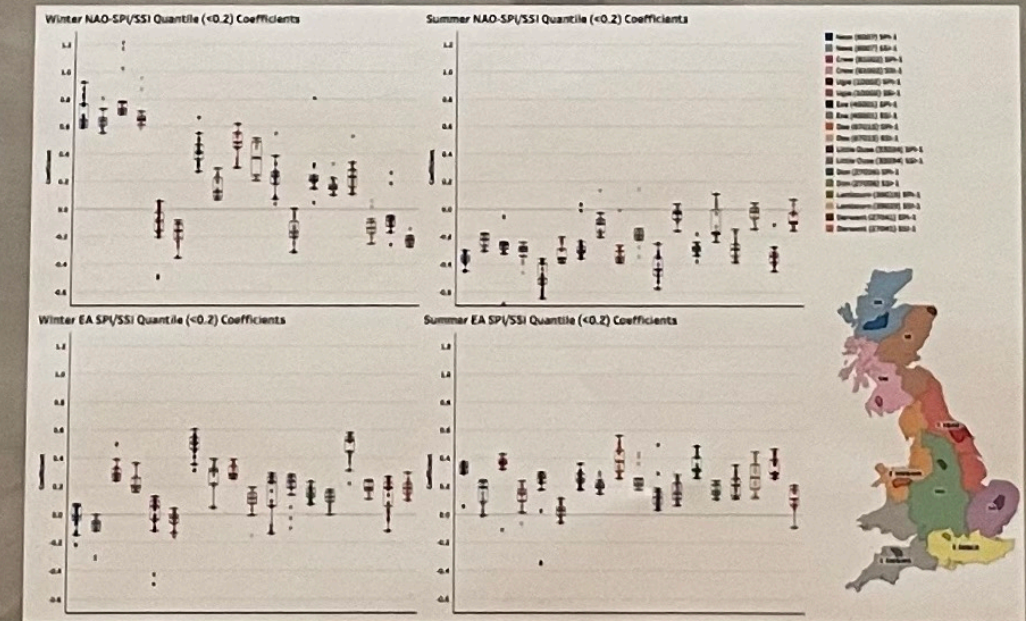
To develop a more detailed understanding of the combined effect of the NAO and EA on meteorological drought, and how rainfall deficits propagate through to hydrological drought, nine case study catchments were selected for further analysis. One catchment from each of the Met Office Climate Districts was chosen to ensure a spatially representative sample across Great Britain. The catchments were chosen as they vary not only geographically, but also represent a range of characteristics such as size, terrain and geology.

Correlations for the full 291 catchments can be found in West et al. (2022a). The graph below shows the correlation coefficients for the nine case study catchments for each month. Quantile regression analysis and a drought severity/frequency analysis was then undertaken to explore the relationship between the two circulations and low/negative SPI-/SSI-I values.

2a. CORRELATION ANALYSIS



2b. QUANTILE REGRESSION



The correlation and QR analysis shows:

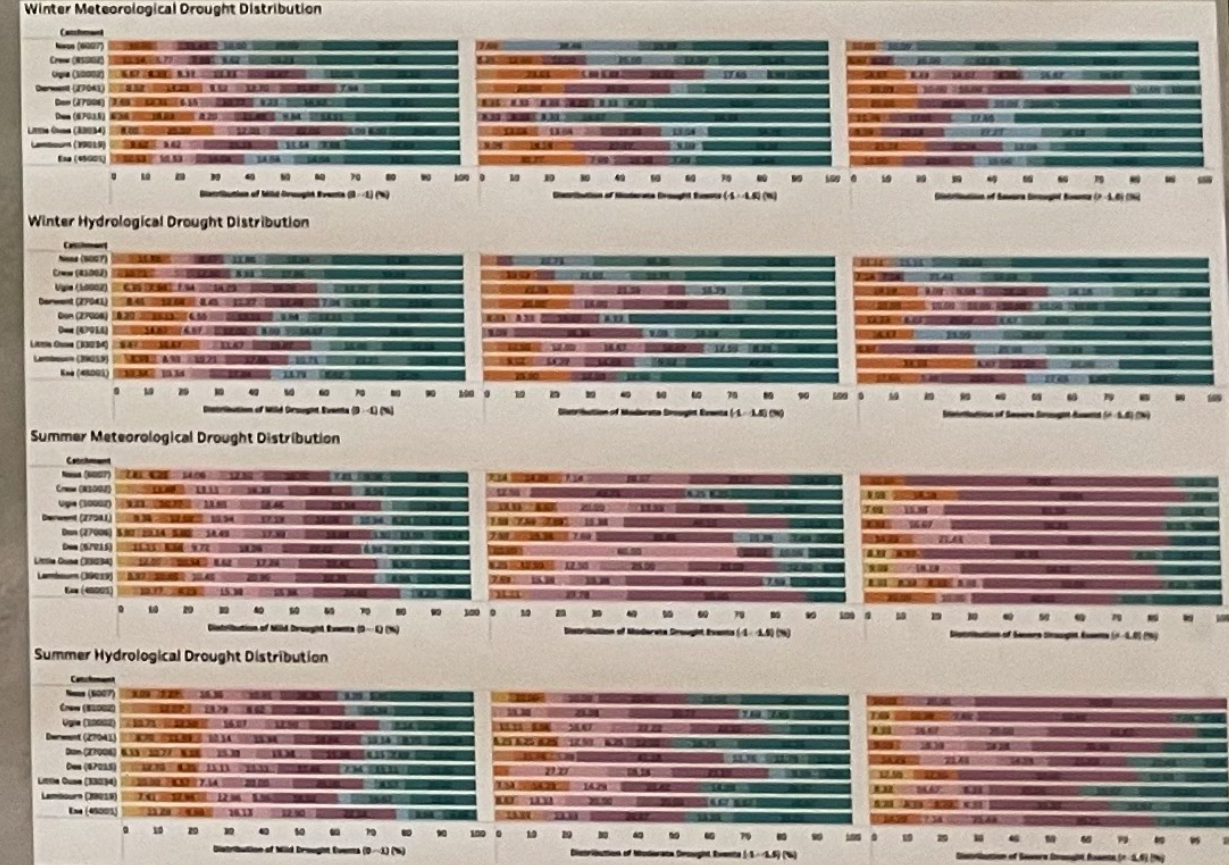
- Catchments in the NW have strong winter low rainfall/flow with the NAO. Whilst EA associations are stronger in the southern catchments.
- In summer more spatially consistent (and somewhat weaker) associations are present.
- Catchments variably show circulation-rainfall-flow propagation due to certain characteristics (e.g. high bedrock permeability in some case study catchments) (West et al 2022c).

2c. DROUGHT FREQUENCY ANALYSIS

A frequency analysis quantifying the relationship between the phase of the two teleconnections and low SPI-I and SSI-I values was undertaken. We also calculated the frequency of mild (0 > Index < -1), moderate (-1 > Index < -1.5) and severe (Index < -1.5) drought under different combinations of NAO and EA phases. The monthly teleconnection index values were classified. Months with teleconnection index values > 0.25 were classified as positive phases, whilst months with index values < -0.25 were classified as negative phases.

This analysis shows:

- In the NW during winter meteorological and hydrological droughts are associated with NAO- conditions. Southern catchments more variably have drought associated with EA- conditions.
- In the summer NAO+ conditions result in drought conditions, however the severity of the drought is potentially moderated/enhanced by the phase of the EA.



3. SUMMARY

Through a convergence of evidence from these analyses we make three key observations:

- Firstly, in the winter months both the NAO and EA exert an influence on drought conditions, however there is spatial variability in the relative influence of the NAO and EA: the NAO has a stronger influence in the north-west, whilst the EA has a stronger influence in the southern and central regions.
- Secondly, in the summer months, less distinctive spatial differences were found, with higher probability of drought conditions under NAO+ phases, which however can be enhanced or moderated by the EA.
- Finally, as a result of catchment characteristics there is spatio-temporal variability in the propagation of meteorological to hydrological drought.

Our findings suggest that by considering the NAO and EA in combination, we can better describe climate and drought variability in both the winter and summer months.

Amini, M., Ghadami, M., Fathian, F. & Modarres, R. (2020). Teleconnections between oceanic-atmospheric indices over Iran using quantile regressions. *Hydrological Sciences Journal*, Vol.65, 2286-2295.
Barker, L.J., Hannaford, J., Osier, A. & Swenson, C. (2018). From meteorological to hydrological drought using standardised indicators. *Hydrology and Earth System Sciences*, Vol.20, 2483-2505.
Barker, L.J., Smith, K.A., Swenson, C., Tanguy, M. & Hannaford, J. (2018). Historic Standardised Streamflow Index (SSI) using Twentieth century distribution with standard period 1961-2010 for 303 UK catchments (1891-2015). NERC Environmental Information Data Centre. <https://doi.org/10.5285/58ef13a0-539f-46e3-884d-c8224799f70>.
Comas-Bru, L. & McDermott, F. (2014). Impacts of the EA and SCA patterns on European twentieth century NAO-winter climate relationship. *Quarterly Journal of the Royal Meteorological Society*, Vol.140, 354-363.
Hurrell, J.W., Kushnir, Y., Ottensm, F. & Visbeck, M. (2003). An overview of the North Atlantic Oscillation. In Hurrell, J.W., Kushnir, Y., Ottensm, F. & Visbeck, M. (eds.), *The North Atlantic Oscillation: Climate Significance and Environmental Impact*. AGU Geophysical Monograph Series, Vol.134, American Geophysical Union: Washington.
Moore, G.W.K., Renfrew, I.A. & Pickett, R.S. (2013). Multidecadal variability of the North Atlantic Oscillation. *Journal of Climate*, Vol.26(8), 2453-2466.
Tanguy, M., Fry, M., Swenson, C. & Hannaford, J. (2017). Historic gridded Standardized Precipitation Index for the United Kingdom 1862-2015 generated using gamma distribution with standard period 1961-2010 u4. NERC Environmental Information Centre. <https://doi.org/10.5285/23090602-4514-4e10-9f9c-9923a2350f7>.
Van Loon, A.F. & Laaha, G. (2015). Hydrological drought severity explained by climate and catchment characteristics. *Journal of Hydrology*, Vol.526, 3-14.
West, H., Quinn, N. & Horswell, M. (2021). Regional Rainfall Response to the North Atlantic Oscillation in Great Britain. *Hydrology Research*, Vol.5(2), 1549-1563.
West, H., Quinn, N. & Horswell, M. (2021). Monthly Rainfall Signatures of the North Atlantic Oscillation & East Atlantic Pattern in Great Britain. *Atmosphere*, Vol.12(11), 1833.
West, H., Quinn, N. & Horswell, M. (2022a). The influence of the North Atlantic Oscillation & East Atlantic Pattern on Drought in British Catchments. *Frontiers in Environmental Science*, Vol.10, 754597.

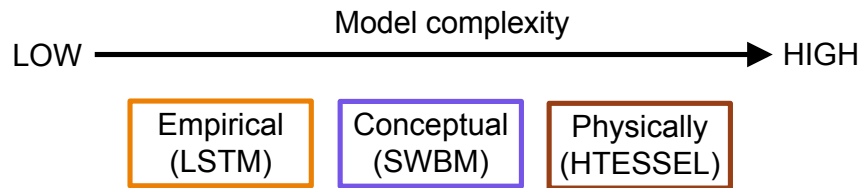
Example 3

Getting the best of both worlds: physics-guided machine learning

Sungmin O and Rene Orth
Hydrology-Biosphere-Climate Interactions

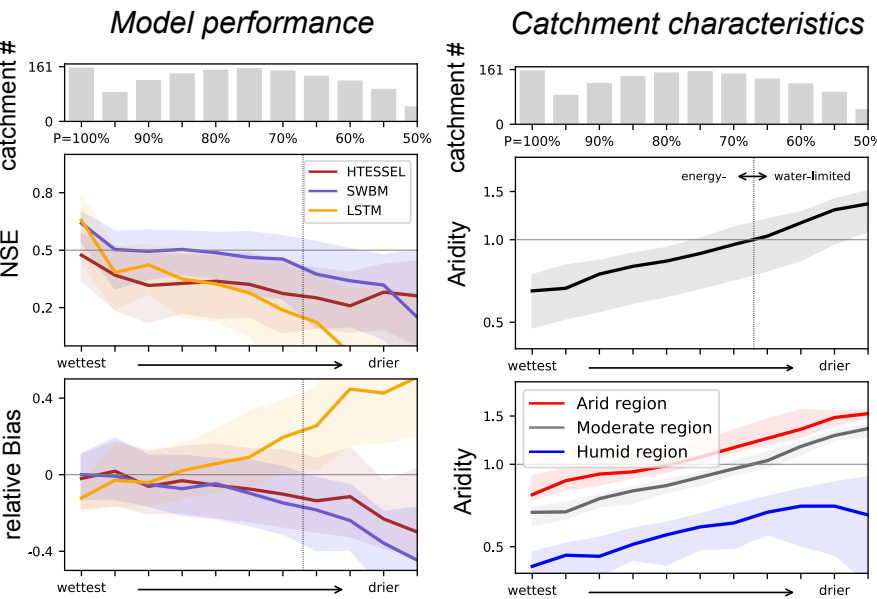


Q1. How reliable are land surface models (LSMs) under a changing climate?



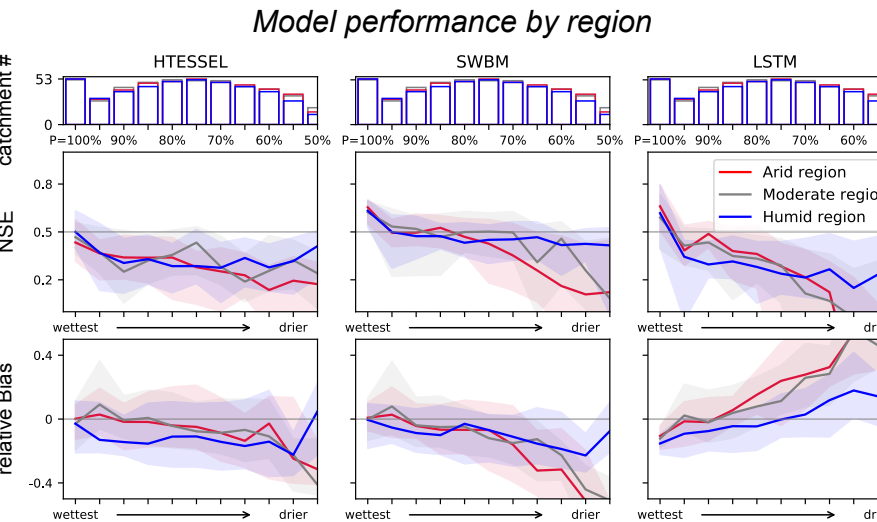
Calibration to the wettest year vs. Evaluation over the remaining years using 24-years streamflow data from 161 catchments in Europe

1. Process-based models show comparatively robust performance under changing conditions



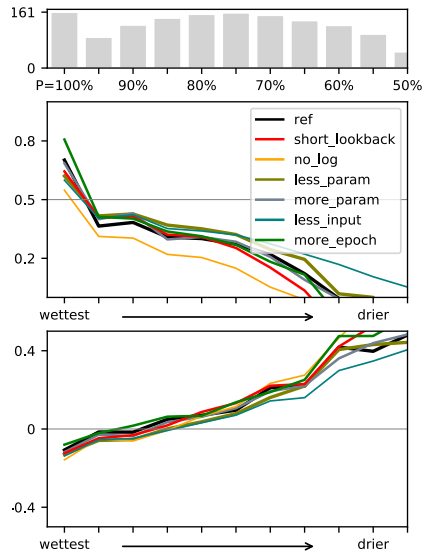
- Models have difficulties in representing hydro-climatic regime shifts; e.g., energy- to water-limited conditions
- Relatively robust performance of the most complex model, HTESSEL, highlights the need for further improved and expanded process representations in current models.

2. Models show faster performance decline in arid regions



- Potential difficulties in LSM predictions are found in semi-arid regions, where a regime change between energy- and water-limited conditions is expected.

Q2. How can machine-learning techniques assist hydro-meteorological forecasting?



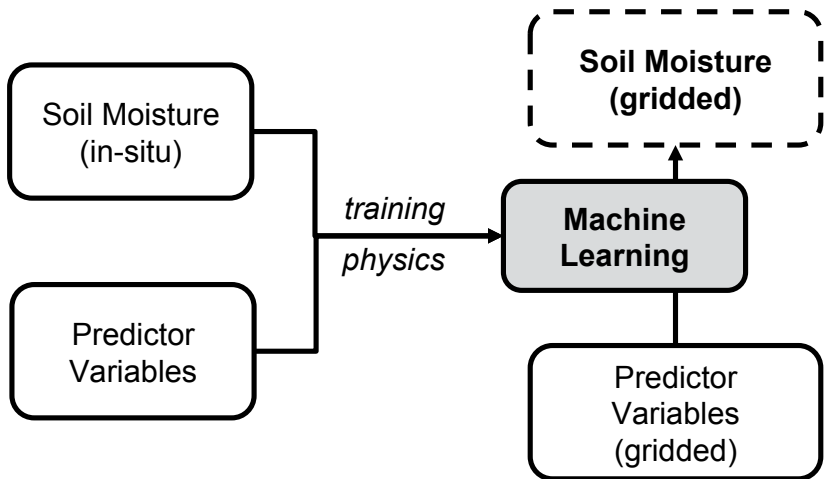
Long Short-Term Memory (LSTM)

- General conclusions remain the same, however, certain hyperparameters play a relatively larger role in determining the robustness of model performance

=> **what is the optimal model set-ups for specific hydrologic applications?**

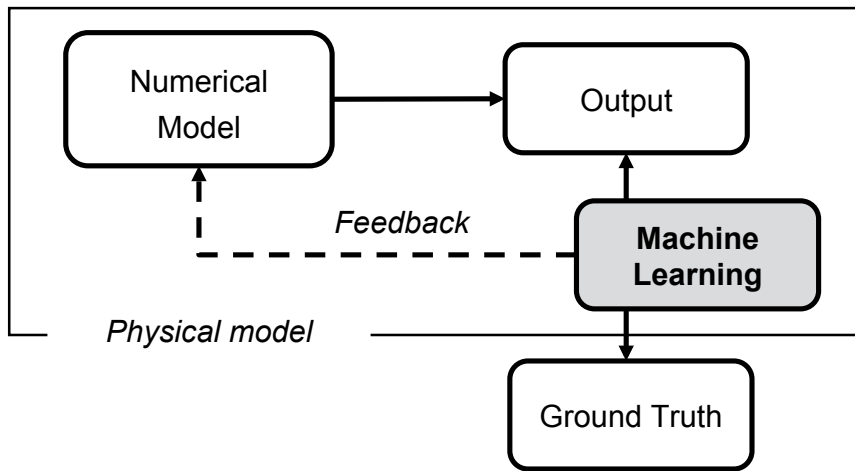
Work Plan 1: Generating gridded soil moisture data from a physic-guided data-driven model

- A data-driven model will be developed and its extrapolation capacity will be investigated, to generate spatio-temporal soil moisture fields from point measurements.

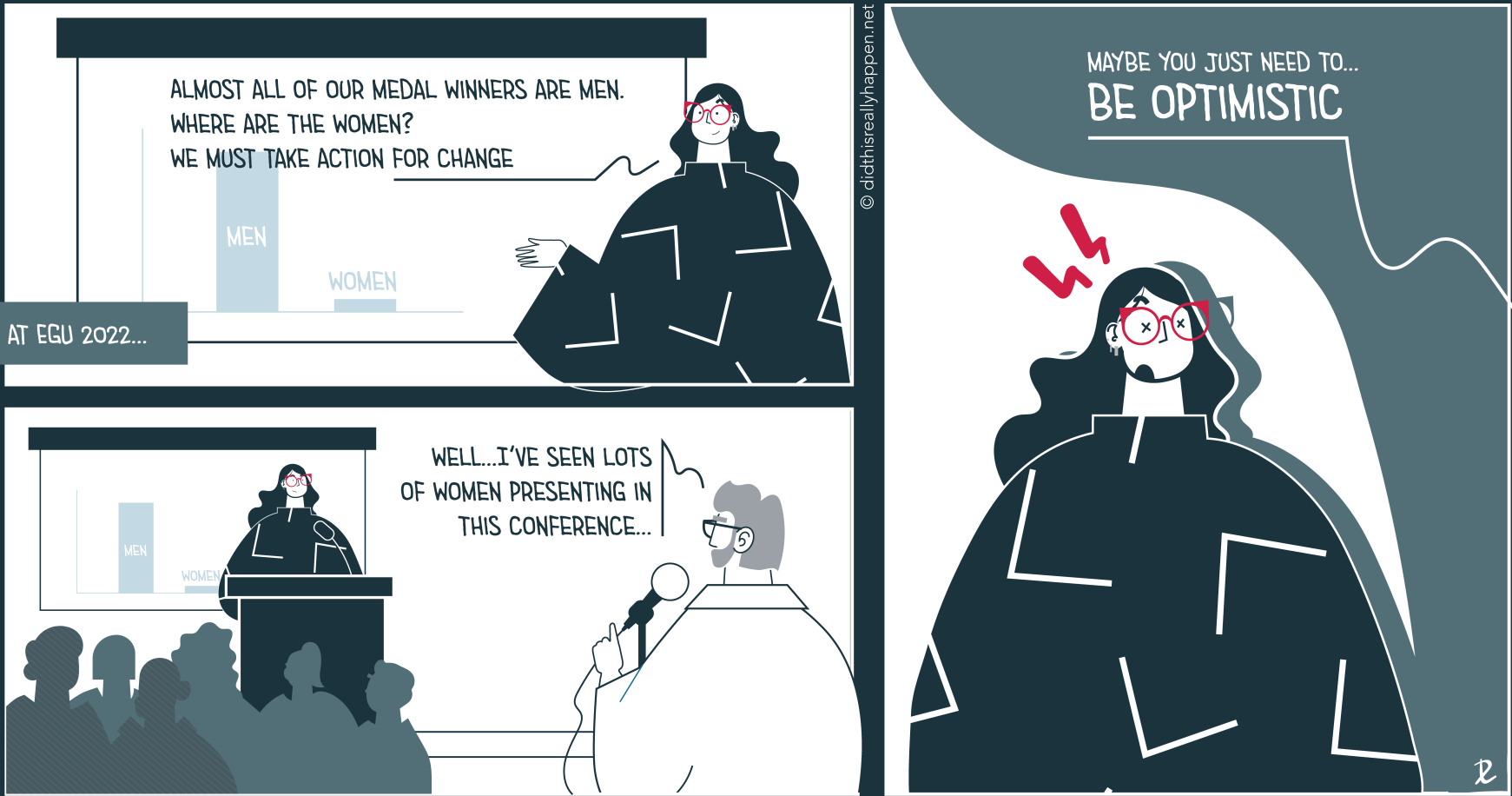


Work Plan 2: Post-processing of weather forecasts with machine-learning techniques

- Machine-Learning can learn patterns of model-observation mismatch and correct biases in physical model outputs.
- It can also improve understanding of model weaknesses, e.g., through sensitivity analysis to determine the patterns in model errors.



Example 4



Did this REALLY happen?!
Yes, it really did.
And it still does.
And it will continue to...

...until we **STOP** it.



Is this **really** still happening?!

Kirstie Wright, Claire Mallard, Lucia Perez-Díaz, Maëlis Arnould, and Nicolas Coltice

Fighting against everyday sexism in science,
one comic strip at a time

The "Did this REALLY happen?!" project aims to highlight and fight against everyday sexism in science through comics based on **real** testimonies submitted anonymously by members of the scientific community.

The Did This Really Happen?! project has been operating for over 7 years, documenting the real lived experiences of sexism in scientific environments, including (un)conscious biases and a range of micro- and macro- aggressions. It is run by an international team of earth scientists (Fig. 1), composed of 4 women and 1 man with diverse knowledge of geoscience and having experienced and witnessed gendered stereotypes throughout their careers.

Since 2017, we have received over **280** testimonies which have been turned into over **50** comics by 2 illustrators. We share our work via our website and social media platforms, articles and invited talks. Our project receives submissions from all career levels across academia and within industry, with the majority being early to mid career scientists (Fig 2). While women represent **95%** of our contributors (the remaining **4%** are men and **1%** prefer not to say), we know that sexism affects everyone and we encourage testimonies from all impacted, to help advocate for greater awareness and to take an active role in fighting against sexism in the workplace.

DID IT HAPPEN
TO YOU TOO?

SUBMIT YOUR STORY
THROUGH OUR WEBSITE



Fig.1: The DTRH team, in cartoon form.
Left to right: Claire, Nico, Maëlis, Lucia and Kirstie

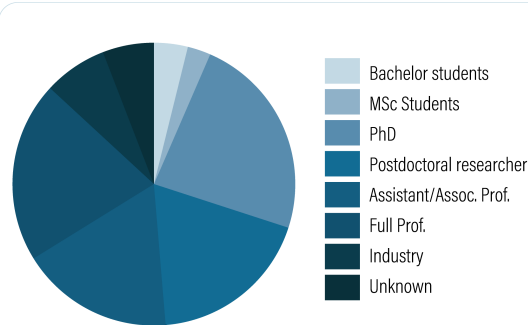


Fig.2: Breakdown of career levels across the anonymous testimonies received between 2017-2022.

Our project uses six non-exclusive categories of sexist biases that have been defined based on the testimonies we receive (Fig. 3). They range from treating women as objects to questioning female competencies and confining males to stereotypical roles (Bocher et al., 2020).

The data we have gathered suggests these potential emerging trends:

- Increasing numbers of testimonies concerning women as outsiders
- Decreasing numbers of testimonies that question female competence
- A relative consistent number of testimonies regarding the objectification of women

The remaining categories show no trends, with the numbers of testimonies received fluctuating from year to year.

While our sample size is small, every data point is a story of everyday sexism in the work place and the impact on that person is very real.

The continued submission of testimonies and the reoccurring sexist situations in which they can be categorised suggests that such behaviours are still widespread. Through comics, we have created an approachable method of communication to show challenging situations without the need for explanations or interpretations.

While our short term goal is to continue to document sexist situations through comics, our longer term aim is to expand the scope of our project, through engagement with other voices and collaboration with organisations to drive change.

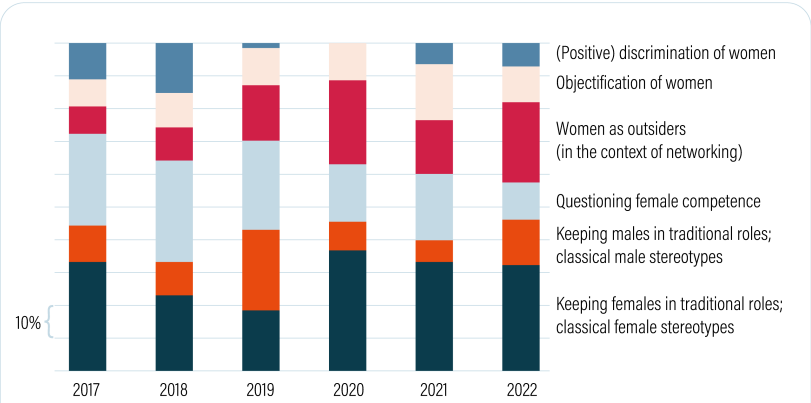


Fig.3: Normalised annual breakdown of stereotypes received through anonymous testimonies.

@TEAM_DTRH
WWW.DIDTHISREALLYHAPPEN.NET



Get involved:

Support our work by engaging with us on social media and sharing our comics!

Example 5



Quantifying CO₂ fertilization effect on GPP from FLUXNET observations

Chunhui Zhan^{1,3*}, René Orth¹, Markus Reichstein¹, Sönke Zaehle¹, Mirco Migliavacca², Anja Rammig³ and Alexander J. Winkler¹

Introduction

- ❑ **CO₂ fertilization effect:** Plant responses to increasing atmospheric CO₂ lead to increasing terrestrial ecosystem carbon storage, thereby reducing the atmospheric CO₂ growth^[1];
- ❑ **Disentangling the CO₂ fertilization effect (β) on gross primary production (GPP)** from other controls such as climatic variability or nitrogen deposition is challenging when using real-world data;
- ❑ We develop a **statistical method** to isolate the CO₂ fertilization effect on GPP and validate the method using a terrestrial ecosystem model.

Take home messages

- ❑ We isolate the CO₂ fertilization effect on GPP (β) by removing the climate impact using a statistical method;
- ❑ The method is overall successfully validated with QUINCY simulations, while some overestimations of β are found in the summer months;
- ❑ Averaging β across many sites is likely to allow approaching a more representative value as site-specific influences partly cancel out.

Goal: Isolating CO₂ fertilization effect on GPP by removing climate effects learned at short time scales

Method

1. Detrend time series

- Calculate monthly mean value (e.g., July in 1990....., 2018) for all variables
- Remove the long-term trend by linear-fit
- Re-scale anomalies to the mean of the original time series



Detrending allows to separate the GPP variability at long time scales (i.e., the trend) which is driven by both CO₂ and climate from that at shorter time scales which is driven by climate only

2. Build model for GPP_detrended = f (climate_detrended)

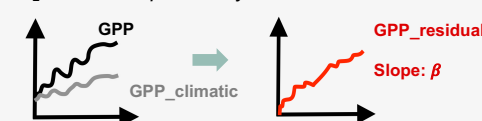
- Using random forest and multivariate linear regression
- Predictors (all detrended): Temperature, VPD, Shortwave radiation, Soil moisture
- Target: detrended GPP

3. Apply model to estimate the effect of climate variability and trends on GPP

- Predictors: Temperature, VPD, Shortwave radiation, Soil moisture
- Result: GPP_climatic

4. Estimate non-climatic effects on GPP

- GPP_residual = GPP - GPP_climatic
 - $\beta = \Delta \text{GPP_residual} / \Delta \text{CO}_2$
- β includes the CO₂ effects but potentially also other non-climatic effects



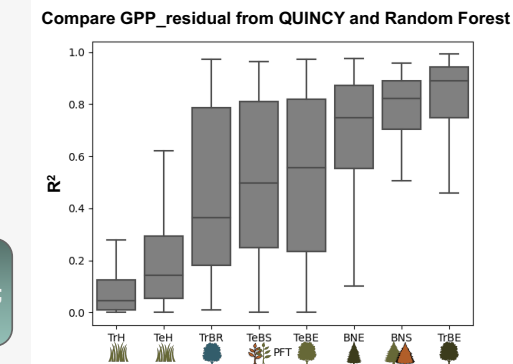
Outlook

- ❑ Include variables related to phenology as a predictor;
- ❑ Validate the method with other models (e.g. TRENDY);
- ❑ Estimate the sensitivity of GPP to climatic variations (γ);
- ❑ Apply statistical method to GPP signals of different time scales determined through Fourier decomposition.

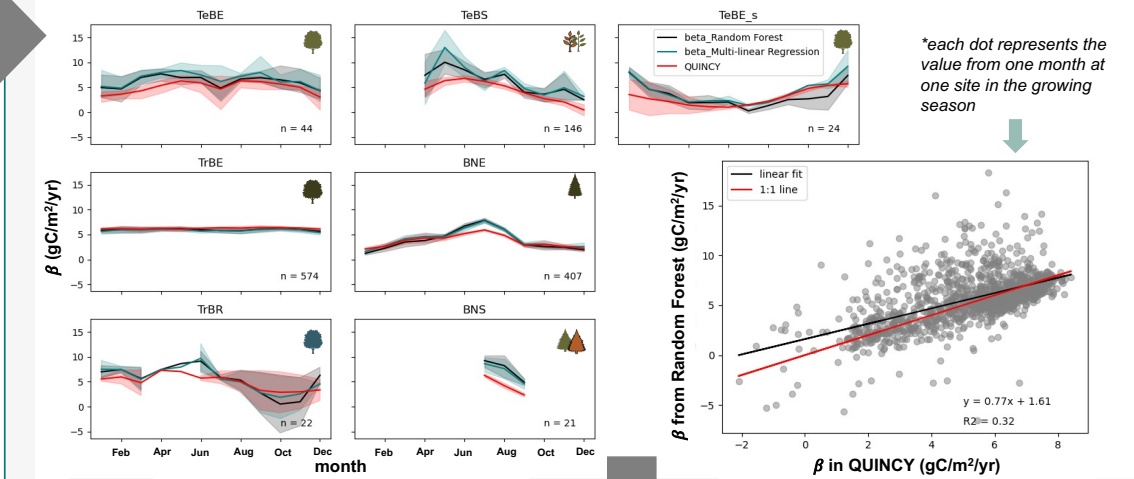
Validate statistical methods with QUINCY simulations

- Using the terrestrial biosphere model QUINCY^[2] we perform two simulations with identical climate but varying CO₂ concentrations (transient CO₂ as observed, and constant CO₂ at levels of year 1901) at several sites distributed across the globe.
- The difference between the GPP simulated in both simulations represents GPP_residual.

- The method works well for tropical forest.
- The method is not valid for grassland sites (TrH & TeH);
- The method overestimates β in summer months;

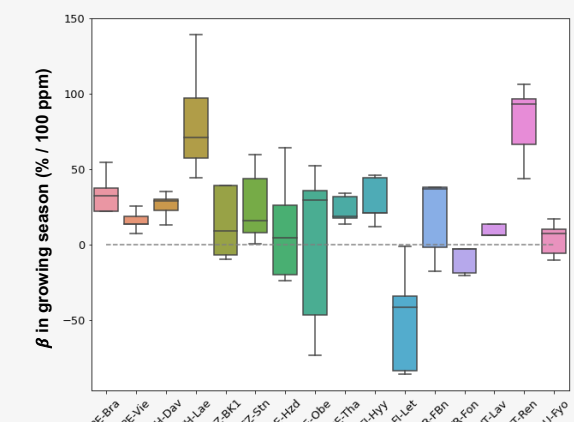
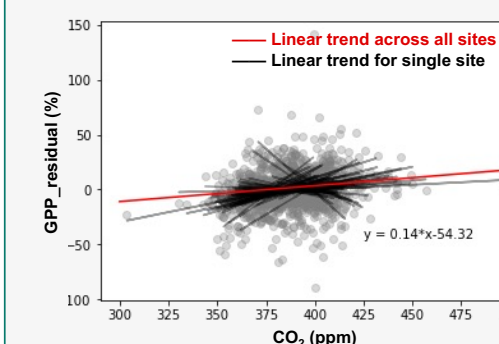


Compare β from QUINCY and statlcal methods



Apply statistical method to FLUXNET observations

- 16 forested sites from ICOS Warmwinter 2020
- Time period >= 10 years



- The average estimated β across 16 forested sites is 14%/100ppm;
- Strong variability across sites may relate to vegetation dynamics.



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References:
[1] Walker, A.P. et al., 2021. Integrating the evidence for a terrestrial carbon sink caused by increasing atmospheric CO₂. New Phytol. 229, 2413–2445.
[2] Thum, T., et al., 2019. A new model of the coupled carbon, nitrogen, and phosphorus cycles in the terrestrial biosphere (QUINCY v1.0; revision 1996). Geosci. Model Dev. 12, 4781–4802.



Consultations



- One consultation is mandatory.
- Best, if you come prepared. Consultation is about giving feedback, inducing ideas and showing opportunities. If there is nothing, I can't help anything.
- Remember that not everybody can come in the last week before the presentation.
- Don't hesitate to e-mail me and ask questions.

Evaluation criteria

	Points
Format/Readability	
Layout/Font size and type/Figure labels	3
Clarity and separability of sections	2
Originality/Catchyness	1
Content	
Description research focus	3
Introduction: importance/motivation/state-of-the-art for the topic; match with research focus	2
Methods: Explanation of important terminology, analysis, used data	2
Results: Figures support findings; Results are comprehensive & match research focus	3
Discussion: Take home messages und lessons learned about research focus	2
Limitations of results	1
References: Continuous reference style; information is referenced	1
Total	20

How and when to print the posters?

- Every body prints posters themselves in the printing station of the university and has them ready on the poster presentation date 14.02.2024.
- We organise poster walls and pins for hanging up your posters.
- The printing station is informed that you come for printing on the 12th February.
- At the printing station mention the module, the professor as well as the Kostenstelle 11000200211.

**Thank you for your attention.
Feel free to ask questions.**

Links for the used images

- Conference: <https://www.google.com/url?sa=i&url=https%3A%2F%2Fblogs.egu.eu%2Fgeolog%2Fcategory%2Fgeneral-assembly%2Fsessions%2Fposters%2F&psig=AOvVaw226cs77XFv0YreBygoUnxN&ust=1697271076954000&source=images&cd=vfe&opi=89978449&ved=0CBQQ3YkBahcKEwjIzdnTyfKBaxUAAAAAHQAAAAAQAw>
- Wordart: https://www.google.com/url?sa=i&url=https%3A%2F%2Ftwitter.com%2Fsketchscience%2Fstatus%2F1376984195277422593&psig=AOvVaw3wf_qrHG4UHLFBNLgx8_cJ&ust=1697271990704000&source=images&cd=vfe&opi=89978449&ved=0CBQQ3YkBahcKEwioi_6HzfKBaxUAAAAAHQAAAAAQHg
- Cat: https://www.google.com/url?sa=i&url=https%3A%2F%2Fcats.com%2Fcat-tail-meanings&psig=AOvVaw1yJDyJWGNQeqOzxz_BXeCu&ust=1697275172209000&source=images&cd=vfe&opi=89978449&ved=0CBQQ3YkBahcKEwjw0sP62PKBaxUAAAAAHQAAAAAQAw
- Burn retina: <https://blogs.lse.ac.uk/impactofsocialsciences/2018/05/11/how-to-design-an-award-winning-conference-poster/>
- Paper on poster: https://www.google.com/url?sa=i&url=https%3A%2F%2Fwww.animateyour.science%2Fpost%2Fbest-examples-of-scientific-posters&psig=AOvVaw3wf_qrHG4UHLFBNLgx8_cJ&ust=1697271990704000&source=images&cd=vfe&opi=89978449&ved=0CBQQ3YkBahcKEwioi_6HzfKBaxUAAAAAHQAAAAAQGg
- It's a match: https://www.google.com/url?sa=i&url=https%3A%2F%2Fwww.reddit.com%2Fr%2FMemeTemplatesOfficial%2Fcomments%2F83wijs%2Ftinder_its_a_match_template%2F&psig=AOvVaw0bZv6lq5uexHR8IsIMeD5-&ust=1697275038171000&source=images&cd=vfe&opi=89978449&ved=0CBQQ3YkBahgKEwigg_42fKBaxUAAAAAHQAAAAAQygE
- Poster transformation: https://www.google.com/url?sa=i&url=https%3A%2F%2Fwww.animateyour.science%2Fpost%2Fbest-examples-of-scientific-posters&psig=AOvVaw0w_87xzDYYdl3x6d7GGmva&ust=1697277708313000&source=images&cd=vfe&opi=89978449&ved=0CBQQ3YkBahcKEwjwvemo4vKBaxUAAAAAHQAAAAAQBQ
- Dog it's not a match: <https://www.google.com/url?sa=i&url=https%3A%2F%2Fwww.britannica.com%2Fanimal%2Fdog&psig=AOvVaw05uajhUQ74MupZH7RgxFjY&ust=1697275479237000&source=images&cd=vfe&opi=89978449&ved=0CBEQjRxqFwoTCMjumYja8oEDFQAAAAAdAAAAABAD>
- Boromir: https://www.google.com/url?sa=i&url=https%3A%2F%2Fsciencetelling.wordpress.ncsu.edu%2F2020%2F02%2F25%2Fspring-2020-meeting-2-some-guidelines-for-research-poster-design%2F&psig=AOvVaw0TCQzfJUCMaLxsv_CN1sRh&ust=1697270117438000&source=images&cd=vfe&opi=89978449&ved=0CBQQ3YkBahcKEwj4IMWGXvKBaxUAAAAAHQAAAAAQDg
- Scatterplot: <https://www.google.com/url?sa=i&url=https%3A%2F%2Fwww.datawrapper.de%2Fcharts%2Fscatter-plot&psig=AOvVaw1V5ayTC0t3-OZOKyioL5ly&ust=1697282624019000&source=images&cd=vfe&opi=89978449&ved=0CBEQjRxqFwoTCKD089b08oEDFQAAAAAdAAAAABAD>